

Big-M method

The Big M method is used to solve linear programming problems with greater-than or equal constraints because it extends the simplex algorithm to these types of problems:

Big-M method

In the previous article of the simplex algorithm we have seen that the method must start with a basic feasible solution. In previous examples so far, we have looked at problems that, when put into standard LP form, conveniently have an all slack starting solution. An all slack solution is only a possibility when all of the constraints in the problem have inequalities. Today, we are going to look at methods for dealing with LPs having other constraint types.

Remember that simplex needs a place to start. It must start from a basic feasible solution then move to another basic feasible solution to improve the objective value. With these assumptions, we can obtain an initial basic feasible solution/dictionary by letting all slack variables be basic, all original variables be non-basic. Obviously, these assumptions do not hold for every LP. What do we do when they don't?

When a basic feasible solution is not readily apparent, the Big M method or the two-phase simplex method may be used to solve the problem.

Steps of Big-M method of LPP

Step-1: At first we have to express the a Linear Programming Problem (LPP) to its standard form.

Step-2: Add artificial variable R_i to the left hand side of all constraints of the form $=$ or $\geq$ types of original LP.

Step-3: We continue with the regular steps of simplex method.

Step-4: While making iterations using simplex method, one of the following cases may arise-

Case-i: If no artificial variable remains in the positive levels and the optimality condition is satisfied, then the solution is optimal.

Case-ii: When big M simplex method terminates with an optimum table, it is sometimes possible for one or more artificial variables to remain in the basis as basic variables. It implies that the original problem is infeasible.

Use penalty (Big-M) method to solve the following LP problem

$$\min \quad Z = x_1 + x_2 + x_3$$

$$x_1 + 2x_2 \geq 3$$

$$x_2 + 7x_3 \leq 6$$

$$x_1 - x_2 + 5x_3 = 5$$

$$x_1, x_2, x_3 \geq 0$$

Slack variable x_5 and surplus variable x_4

$$\max \quad Z = -x_1 - x_2 - x_3$$

Such that

$$x_1 + 2x_2 - x_4 = 3$$

$$x_2 + 7x_3 + x_5 = 6$$

$$x_1 - x_2 + 5x_3 = 5$$

$$x_1, x_2, x_3 \geq 0$$

Now add two variable (*artificial variables*) x_6 and x_7 costcoefficient very high .
Now problem reduce to

$$\max \quad Z = -x_1 - x_2 - x_3 + 0x_4 + 0x_5 - M(x_6 + x_7)$$

Such that

$$x_1 + 2x_2 + 0x_3 - x_4 + 0x_5 + x_6 + 0x_7 = 3$$

$$0x_1 + x_2 + 7x_3 + 0x_4 + x_5 + 0x_6 + 0x_7 = 6$$

$$x_1 - x_2 + 5x_3 + 0x_4 + 0x_5 + 0x_6 + x_7 = 5$$

	C_j	-1	-1	-1	0	0	-M	-M	
C_B	B_V	x_1	x_2	x_3	x_4	x_5	x_6	x_7	X_B
-M	x_6	1	2	0	-1	0	1	0	3
0	x_5	0	1	7	0	1	0	0	6
-M	x_7	1	-3	5	0	0	0	1	5
	Z_j	-2M	M	-5M	M	0	-M	-M	
	$Z_j - C_j$								

	C_j	-1	-1	-1	0	0	-M	-M	
C_B	B_V	x_1	x_2	x_3	x_4	x_5	x_6	x_7	X_B
-M	x_6	1	2	0	-1	0	1	0	3
0	x_5	0	1	7	0	1	0	0	6
-M	x_7	1	-3	5	0	0	0	1	5
	Z_j	-2M	M	-5M	M	0	-M	-M	
	$Z_j - C_j$	-2M+1	M+1	-5M+1	M	0	0	0	

	C_j	-1	-1	-1	0	0	-M	-M	
C_B	B_V	x_1	x_2	x_3	x_4	x_5	x_6	x_7	X_B
-M	x_6	1	2	0	-1	0	1	0	3
0	x_5	0	1	7	0	1	0	0	6
-M	x_7	1	-3	5	0	0	0	1	5
	Z_j	-2M	M	-5M	M	0	-M	-M	
	$Z_j - C_j$	-2M+1	M+1	-5M+1	M	0	0	0	

minimum $\{-2M+1, M+1, -5M+1, M\} = -5M+1$

x_3 is entering basis

Find the row column

	C_j	-1	-1	-1	0	0	-M	-M	
C_B	B_V	x_1	x_2	x_3	x_4	x_5	x_6	x_7	X_B
-M	x_6	1	2	0	-1	0	1	0	3
0	x_5	0	1	7	0	1	0	0	6
-M	x_7	1	-3	5	0	0	0	1	5
	Z_j	-2M	M	-5M	M	0	-M	-M	
	$Z_j - C_j$	-2M+1	M+1	-5M+1	M	0	0	0	

$$\text{Minimum}\left\{\frac{3}{0}, \frac{6}{7}, \frac{5}{5}\right\} = \frac{6}{7}$$

x_5 is leaving basis

Find the row column

	C_j	-1	-1	-1	0	-M	-M	
C_B	B_V	x_1	x_2	x_3	x_4	x_6	x_7	X_B
-M	x_6	1	2	0	-1	1	0	3
-1	x_3	0	1	7	0	0	0	6
-M	x_7	1	-3	5	0	0	1	5
	Z_j							
	Z_j-C_j							

$$R'_1 = R_1, R'_2 = \frac{1}{7}R_2, R'_3 = R_3 - \frac{5}{7}R_2,$$

Find the row column

	C_j	-1	-1	-1	0	-M	-M	
C_B	B_V	x_1	x_2	x_3	x_4	x_6	x_7	X_B
-M	x_6	1	2	0	-1	1	0	3
-1	x_3	0	$\frac{1}{7}$	1	0	0	0	$\frac{6}{7}$
-M	x_7	1	$\frac{-26}{7}$	0	0	0	1	$\frac{5}{7}$
	Z_j							
	Z_j-C_j							

Find the row column

	C_j	-1	-1	-1	0	-M	-M	
C_B	B_V	x_1	x_2	x_3	x_4	x_6	x_7	X_B
-M	x_6	1	2	0	-1	1	0	3
-1	x_3	0	$\frac{1}{7}$	1	0	0	0	$\frac{6}{7}$
-M	x_7	1	$\frac{-26}{7}$	0	0	0	1	$\frac{5}{7}$
	Z_j	-2M	$\frac{12M - 1}{7}$	-1	M	-M	-M	
	$Z_j - C_j$	-2M-1	$\frac{12M + 6}{7}$	0	0	0	0	

Find the row column

	C_j	-1	-1	-1	0	-M	-M	
C_B	B_V	x_1	x_2	x_3	x_4	x_6	x_7	X_B
-M	x_6	1	2	0	-1	1	0	3
-1	x_3	0	$\frac{1}{7}$	1	0	0	0	$\frac{6}{7}$
-M	x_7	1	$-\frac{26}{7}$	0	0	0	1	$\frac{5}{7}$
	Z_j	-2M	$\frac{12M-1}{7}$	-1	M	-M	-M	
	Z_j-C_j	-2M-1	$\frac{12M+6}{7}$	0	0	0	0	

minimum $\{-2M-1, \frac{12M+6}{7}\}=-2M-1$

x_1 is entering basis

Find the row column

	C_j	-1	-1	-1	0	-M	-M	
C_B	B_V	x_1	x_2	x_3	x_4	x_6	x_7	X_B
-M	x_6	1	2	0	-1	1	0	3
-1	x_3	0	$\frac{1}{7}$	1	0	0	0	$\frac{6}{7}$
-M	x_7	1	$-\frac{26}{7}$	0	0	0	1	$\frac{5}{7}$
	Z_j							
	$Z_j - C_j$							

Minimum $\{\frac{3}{1}, \frac{6/7}{0}, \frac{5/7}{1}\}=\frac{5/7}{1}$

x_1 is entering basis

Find the row column

	C_j	-1	-1	-1	0	-M	-M	
C_B	B_V	x_1	x_2	x_3	x_4	x_6	x_7	X_B
-M	x_6	1	2	0	-1	1	0	3
-1	x_3	0	$\frac{1}{7}$	1	0	0	0	$\frac{6}{7}$
-M	x_7	1	$-\frac{26}{7}$	0	0	0	1	$\frac{5}{7}$
	Z_j							
	$Z_j - C_j$							

x_7 is leaving basis x_1 is entering basis

Find the row column

	C_j	-1	-1	-1	0	-M	
C_B	B_V	x_1	x_2	x_3	x_4	x_6	X_B
-M	x_6	1	2	0	-1	1	3
-1	x_3	0	$\frac{1}{7}$	1	0	0	$\frac{6}{7}$
-1	x_1	1	$-\frac{26}{7}$	0	0	0	$\frac{5}{7}$
	Z_j						
	$Z_j - C_j$						

$$R'_1 = R_1 - R_3, R'_2 = R_2, R'_3 = R_3$$

Find the row column

	C_j	-1	-1	-1	0	-M	
C_B	B_V	x_1	x_2	x_3	x_4	x_6	X_B
-M	x_6	0	$\frac{40}{7}$	0	-1	1	$\frac{16}{7}$
-1	x_3	0	$\frac{1}{7}$	1	0	0	$\frac{6}{7}$
-1	x_1	1	$-\frac{26}{7}$	0	0	0	$\frac{5}{7}$
	Z_j						
	$Z_j - C_j$						

$$R'_1 = R_1 - R_3, R'_2 = R_2, R'_3 = R_3$$

Find the row column

	C_j	-1	-1	-1	0	-M	
C_B	B_V	x_1	x_2	x_3	x_4	x_6	X_B
-M	x_6	0	$\frac{40}{7}$	0	-1	1	$\frac{16}{7}$
-1	x_3	0	$\frac{1}{7}$	1	0	0	$\frac{6}{7}$
-1	x_1	1	$\frac{-26}{7}$	0	0	0	$\frac{5}{7}$
	Z_j	-1	$\frac{-40M + 25}{7}$	-1	M	-M	
	$Z_j - C_j$	0	$\frac{-40M + 25}{7}$	0	M	0	

Find the row column

	C_j	-1	-1	-1	0	-M	
C_B	B_V	x_1	x_2	x_3	x_4	x_6	X_B
-M	x_6	0	$\frac{40}{7}$	0	-1	1	$\frac{16}{7}$
-1	x_3	0	$\frac{1}{7}$	1	0	0	$\frac{6}{7}$
-1	x_1	1	$\frac{-26}{7}$	0	0	0	$\frac{5}{7}$
	Z_j						
	$Z_j - C_j$						

x_2 is entering basis

Find the row column

	C_j	-1	-1	-1	0	-M	
C_B	B_V	x_1	x_2	x_3	x_4	x_6	X_B
-M	x_6	0	$\frac{40}{7}$	0	-1	1	$\frac{16}{7}$
-1	x_3	0	$\frac{1}{7}$	1	0	0	$\frac{6}{7}$
-1	x_1	1	$\frac{-26}{7}$	0	0	0	$\frac{5}{7}$
	Z_j						
	$Z_j - C_j$						

Minimum $\{\frac{16/7}{40/7}, \frac{6/7}{1/7}, \frac{5/7}{-26/7}\}=\frac{16/7}{40/7}$

x_2 is entering basis

x_6 is leaving basis

Find the row column

	C_j	-1	-1	-1	0	
C_B	B_V	x_1	x_2	x_3	x_4	X_B
-1	x_2	0	$\frac{40}{7}$	0	-1	$\frac{16}{7}$
-1	x_3	0	$\frac{1}{7}$	1	0	$\frac{6}{7}$
-1	x_1	1	$\frac{-26}{7}$	0	0	$\frac{5}{7}$
	Z_j					
	$Z_j - C_j$					

Find the row column

	C_j	-1	-1	-1	0	
C_B	B_V	x_1	x_2	x_3	x_4	X_B
-1	x_2	0	$\frac{40}{7}$	0	-1	$\frac{16}{7}$
-1	x_3	0	$\frac{1}{7}$	1	0	$\frac{6}{7}$
-1	x_1	1	$\frac{-26}{7}$	0	0	$\frac{5}{7}$
	Z_j					
	$Z_j - C_j$					

$$R'_1 = \frac{7}{40}R_1, R'_2 = R_2 - \frac{1}{40}R_1, R'_3 = R_3 + \frac{26}{40}R_1$$

Find the row column

	C_j	-1	-1	-1	0	
C_B	B_V	x_1	x_2	x_3	x_4	X_B
-1	x_2	0	1	0	$-\frac{7}{40}$	$\frac{2}{5}$
-1	x_3	0	0	1	$\frac{1}{40}$	$\frac{4}{5}$
-1	x_1	1	0	0	$-\frac{13}{20}$	$\frac{11}{5}$
	Z_j					
	$Z_j - C_j$					

Find the row column

$$\begin{aligned} \max \quad Z &= -x_1 - x_2 - x_3 \\ &= -\frac{11}{5} - \frac{2}{5} - \frac{4}{5} = -\frac{17}{5} \end{aligned}$$

	C_j	-1	-1	-1	0	
C_B	B_V	x_1	x_2	x_3	x_4	X_B
-1	x_2	0	1	0	$-\frac{7}{40}$	$\frac{2}{5}$
-1	x_3	0	0	1	$\frac{1}{40}$	$\frac{4}{5}$
-1	x_1	1	0	0	$-\frac{13}{20}$	$\frac{11}{5}$
	Z_j	-1	-1	-1	$\frac{4}{5}$	
	$Z_j - C_j$	0	0	0	$\frac{4}{5}$	

$$\min \quad Z = \frac{17}{5}$$

Optimal solution $x_1 = \frac{11}{5}$, $x_2 = \frac{2}{5}$, $x_3 = \frac{4}{5}$ is the optimal solution

Find the row column

	C_j	-1	-1	-1	0	
C_B	B_V	x_1	x_2	x_3	x_4	X_B
-1	x_2	0	1	0	$-\frac{7}{40}$	$\frac{2}{5}$
-1	x_3	0	0	$\frac{1}{40}$	0	$\frac{4}{5}$
-1	x_1	1	0	0	0	$\frac{11}{5}$
	Z_j	-1	-1	$-\frac{1}{40}$	$\frac{7}{40}$	
	$Z_j - C_j$	0	0	$\frac{39}{40}$		